Dæmaskammtur 2

To receive full marks, you must show your work. List your assumptions and justify your reasoning where appropriate.

1. The gravitational potential energy of an object with mass m near the surface of the Earth can be written as
$$U(r) = -G\frac{M_\oplus m}{r}, \tag{1}$$
where $M_\oplus$ is the mass of the Earth and r is the distance of the mass from the centre of the Earth. Assuming the Earth is a sphere with radius $R_\oplus$, use Taylor expansion to show that a change in potential energy can be written as $\Delta U = mgh$ for small values of h, where g is a constant.

2. Einstein's theory of special relativity introduces an equation for the kinetic energy of a particle with mass m_0 moving at the speed v:
$$K = \frac{m_0 c^2}{\sqrt{1-(v/c)^2}} - m_0 c^2. \tag{2}$$
Write down the first two terms in the Taylor series for the kinetic energy, K, in terms of v/c. Show that the first order approximation gives $K = \frac{1}{2}m_0 v^2$.

3. You are standing on top of the Burj Khalifa in Dubai which peaks at an altitude of $h = 830$ m above sea level. Show that the distance to the horizon, i.e. to a point on the Earth that sits just below the horizon can be approximated as $s = \sqrt{2Rh}$ where R is the radius of the Earth (assumed to be spherical). Hint: See figure below. Show that $h/R = \sec\theta - 1$ and calculate the first two terms for $\sec\theta = 1/\cos\theta$.

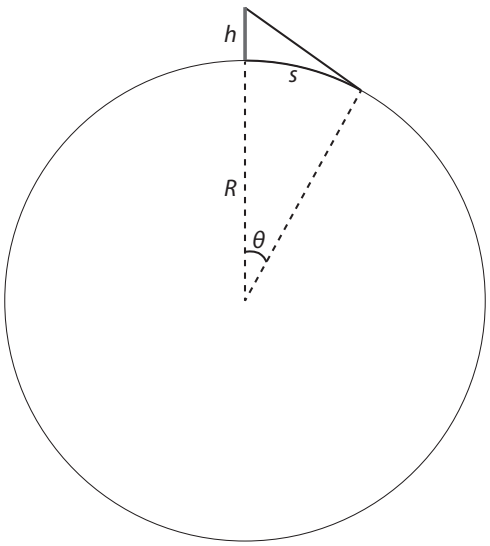


Figure 1: The system described in problem 3.

4. Write a short Python script that creates a figure with four panels representing the function $f(x) = e^x \cos(x)$ in the interval $x \in [-2, 1.7]$ as well as four lines (one in each panel) that indicate $S_n(x)$ for $n = 1, 3, 4, 5$ where $S_n(x)$ is the Taylor series of the function $f(x)$ centred on $c = 0$.

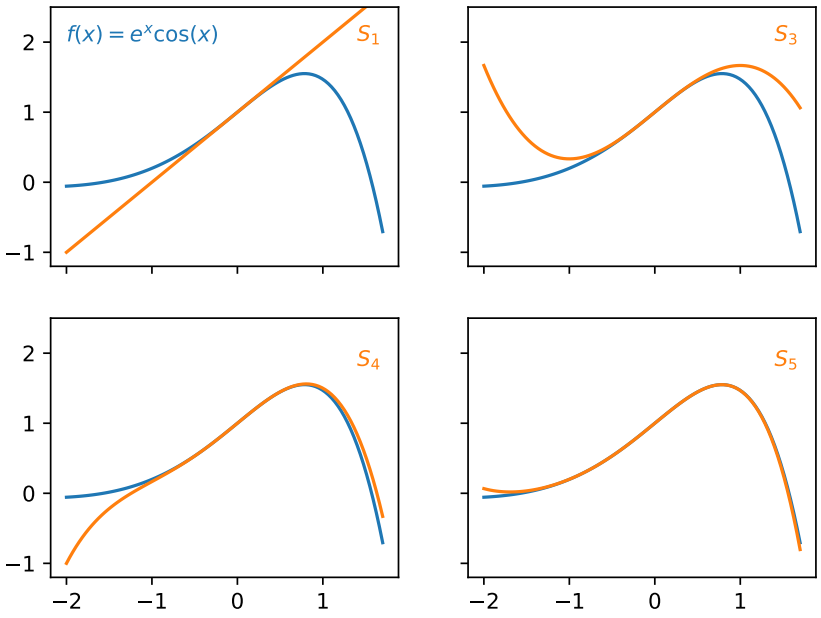


Figure 2: The figure that you are tasked with creating.